INCH-POUND

MIL-P-64008 (AR)
AMENDMENT 4
11 APRIL 1991

SUPERSEDING
AMENDMENT 3
16 May 1984

MILITARY SPECIFICATION

PROJECTILE, 155MM: M825 METAL PARTS ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-64008 (AR), dated 27 December 1982, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Change the title to "PROJECTILE, 155MM: M825/M825A1 METAL PARTS ASSEMBLY".

1.: Delete "cope" and substitute "Scope".

1.1: Delete "XM825" and substitute "M825".

* 1.1: Delete "M825" and replace with "M825/M825A1".

* 2.1.1: Add Military Specification:

"MIL-A-8625 - Anodic coatings, for Aluminum alloys"

* 2.1.1: Add Military Specification:

"MIL-A-70625 - Automated Acceptance Inspection Equipment for Non Electronic Components, Design, Testing and Approval, of"

PAGE 2

* 2.1.1: Specifications and Standards:

Under Product and Packaging drawing 92352634 delete title and substitute "Projectile, 155MM, M825, Metal Parts Assembly".

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2.1.1: Under Inspection Equipment drawings make the following changes:

"T928035 - Eddy Current Standard, XM825 Base" to:

"9280354 - Eddy Current Standard, M825 Base" and

"T9280355 - Envelope drawing Eddy Current Equipment, XM825 Base" to:

"9280355 - Envelope drawing Eddy Current Equipment, M825 Base"

* Change: "9280427 - Envelope Drawing - M825 Base

9280428 - Ultrasonic Standard Drawing, M825 Base"

To: "9280427 - Envelope Drawing - M825A1 Base

9280428 - Ultrasonic Standard Drawing, M825A1 Base".

2.2: Under "American Society for Testing and Material (ASTM)" Add the following:

"ASTM Method E-18-Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials"

* 3.2.3 Heat treatment, Change from:

"Each body (after application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (see 6.12)."

To:

"Each body (after application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. (see 6.12)."

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* Add the following paragraphs:

"3.3.1.1 Minimum melting of steel. Upon removal of the rotating band material, the steel band seat shall demonstrate visual evidence of minimum melting due to gilding metal adhesion during application of the rotating band. Visual comparison standards will be provided by the Design Agency.

3.3.1.2 Subsurface defects. There shall be no evidence of either intergranular penetration of gilding metal or weld start-stop cratering when the body band seat is turned down to a 5.875 plus or minus 0.010 inch diameter."

* 3.3.2: Delete and add the following:

"3.3.2 Rotating band seat. Body material (steel) under the rotating band shall not be visible in the cannelure diameter.

3.3.2.1 Process control. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface. A process requirement that will control this feature is depicted on the Body drawing (see Dwg. 9352632). The contractor may, at his option, use a control other than the one defined on the drawing. The manufacturing procedure and the inspection plan for controlling the process shall be forwarded to the responsible technical agency for approval prior to its implementation into production."

* 3.4: After base add "(9352629)".

* 3.4.1: Delete and substitute the following:

"3.4.1 Microstructure. The microstructure of the ogive and base (9352629) shall comply with the microstructure depicted in Figures 1A and 1B or 2. Two examples of acceptable microstructure are shown in Figure 1A and 1B. In Figure 1A, the insoluble phase is well broken up, randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1B the insoluble phase is well broken up and randomly distributed within a matrix which is partially recrystallized. An example of a marginally acceptable microstructure is shown in Figure 2. In this figure the insoluble second phase exhibits some discontinuity and appears to be elongated. An example of an unacceptable microstructure is shown in Figure 3. The insoluble phase is

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distributed in a more continuous form (Network) within a matrix which shows a large degree of microsegration (or coring). The microstructure in Figure 3 must be avoided because it is detrimental to the integrity of the material."

* 3.4.2: First line, after base add "(9352629)".

* 3.4.2.1: Delete and substitute the following:

"3.4.2.1 Base (9352629) heat treatment verification. Each base (9352629) shall be Rockwell B85 minimum."

* Add paragraph 3.4.2.2:

"3.4.2.2 Base (9378453) heat treatment verification. Each base (9378453) shall be Rockwell C32 minimum."

* 3.4.3: Second line, after base, add "(9352629)".

* Delete paragraph 3.6 in its entirety and substitute the following:

"3.6 Heat treatment of base (9378453) and adapter (9352630). Each base and adapter shall be austenitized, oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing (see 6.12). No work involving plastic flow is permitted after tempering unless followed by a stress relief."

* 3.6 Heat treatment of base (9378453) and adapter (9352630), Change from:

"Each base and adapter shall be austenitized, quenched and tempered uniformily to meet the mechanical properties specified on the applicable drawing (see 6.12). No work involving plastic flow is permitted after tempering unless followed by a stress relief."

To:

"Each base and base adapter shall be austenitized, oil quenched and tempered uniformily to meet the mechanical properties specified on the applicable drawing (see 6.12). No work involving plastic flow is permitted after tempering unless followed by a stress relief."

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* Add paragraph 3.6.1:

"3.6.1 Verification of base (9378453) heat treatment (special defect). After final machining and prior to phosphate coating, each base (9378453) shall be tested for heat treatment. Bases failing to comply will be rejected. All bases passing acceptance will be stamped in accordance with the applicable drawing. Testing shall be in accordance with 4.5.1.3."

PAGE 5

3.7: Delete "XM825" and substitute "M825".

3.8.1: Second line, insert "microstructure," between "process" and "and".

* Add paragraph 3.12 as follows:

"3.12 Protective finish. Details of the proposed procedure for zinc phosphate coating, anodic coating to be used by the contractor shall be submitted in writing to the cognizant engineering agency and written approval received prior to initiation of production. The proposed procedure shall include materials, process and equipment to be used by the contractor, detailed method of control including limits for time, temperature, concentration and all other information pertinent to the processes. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use."

PAGE 6

* 4.3.1: Add the following:

"Base	9378453	26
Base (prior to prime and paint)	9378453	5"

PAGE 7

* Table I. First article inspection. Sheet 1 of 3, delete in its entirety and substitute the following:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER See below
	Metal Parts Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ogive				
	(Dwg. 10542916)				
	Examination for defects	5	-	3.1	4.4.2.1
	Protective finish test	1	-	3.1	4.5.10
	Yield strength & percent elongation	2 Specimens	-	3.4.2	4.5.1.5
	Microstructure	1	-	3.4.1	4.5.7
	Forging	*	-	3.8.1	*
	Obturator				
	(Dwg. 10542907)				
	Examination for defects	5	-	3.1	4.4.2.2
	Base (prior to prime & paint)				
	(Dwg. 9378453)				
	Examination for defects	5		3.1	4.4.2.11
	Metal defective	5		3.5.1	4.4.2.11
	Yield strength & percent elongation	5		3.6	4.4.2.11
	Heat treatment verification	5		3.6.1, 4.5.1.3	4.4.2.11
	Base				
	(Dwg. 9378453)				
	Examination for defects	5		3.1	4.4.2.12
	Salt spray	1		3.1	4.5.10

NOTES: * See appropriate requirement paragraph.

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* Table I. First article inspection. Sheet 2 of 3, delete in its entirety and substitute the following:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Metal Parts Assembly		SHEET 2 OF 3		DRAWING NUMBER: See below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter, Base				
	(Dwg. 9352630)				
	Examination for defects	5	-	3.1	4.4.2.7
	Yield strength & percent elongation	2 Specimens	-	3.1	4.5.1.6
	Base				
	(Dwg. 9352629)				
	Examination for defects	5		3.1	4.4.2.6
	Yield strength & percent elongation	2		3.4.2	4.5.1.5
	Protective finish test	1		3.1	4.5.10
	Microstructure	1		3.4.1	4.5.7
	Forging	*		3.8.1	*
	Metal defective (prior to drilling wrench slots)	5		3.5.1	4.5.4
	Verification of heat treat (prior to protective finish)	5		3.1	4.5.13
	Base and Adapter Assembly				
	(Dwg. 9352631)				
	Examination for defects	5		3.1	4.4.2.8
	Air leak test	5		3.1	4.5.9
	Protective finish test	1		3.1	4.5.10
	Body				
	(Dwg. 9352532)				
	Examination for defects	5	-	3.1	4.4.2.4
	Yield strength & percent elongation	2	-	3.2.3	4.5.1.4

NOTES: * See appropriate requirement paragraph.

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* Delete paragraph 4.4.1, 4.4.1.1 and 4.4.1.2 in their entirety and substitute the following:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Bodies from not more than one forging manufacturer.

b. Ogives from not more than one forging manufacturer.

c. Bases (9378453) from not more than one forging manufacturer.

d. Base and adapter assemblies from not more than one forging manufacturer of each component.

4.4.1.1 Heat treatment lot. A heat treatment lot of bodies, bases (9378453), or base adapters shall consist of bodies, bases or base adapters from one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles in one unchanged process. The maximum lot size shall be 2000. If a mill heat exceeds 2000, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots. The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Bodies or bases in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.2 Miscellaneous heat treatment lots. Bodies, bases or base adapters manufactured from small heats of steel, or that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000."

* 4.4.1.3: Delete and substitute the following:

"4.4.1.3 Heat treatment lot (base, 9352629) and ogive. A heat treatment lot of ogives and bases shall consist of components manufactured from material of the same alloy and temper from one manufacturer. At no time will the lot quantity exceed 4800. A heat treatment lot consisting of a

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single component configuration shall be heat treated by the same continuous process, in the same equipment, and using the same temperature and time cycles. A heat treatment lot shall be considered to have been processed continuously if the parts are solutionized and quenched in an uninterrupted manner with no more than a ten (10) hour delay for non-equipment related shutdown(s) provided normal furnace operation temperatures are maintained. If the heat treat process is interrupted to process different components, different lot numbers must be assigned to the parts produced before and after the interruption. Each heat treat lot shall be aged in the same furnace at the same time."

PAGE 11

* 4.4.1.4: In the title after base add "(9352629)".

4.4.2.a: Add the following to the end of the last sentence:

"or MIL-STD-1235, Inspection Level II where applicable."

PAGE 13

* 4.4.2.1:

Add "(c)" after defect category 118 and 119.

PAGE 14

* 4.4.2.1:

Add the following note to the notes block:

"(c) These inspections are not required when the body and ogive are supplied as an assembled unit."

PAGE 16

* Add the following major defect:

"103	Runout of band seat to bourrelets	4.4.2.d	Gage"

PAGE 17

* 4.4.2.4:

Major 101, Band seat metal defective - change "3.5.2" to "3.3.1, 3.5.2".

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PAGE 19

4.4.2.4:

Major 131: Delete in its entirety.

Major 135: Delete "Gage" and substitute "Visual".

PAGE 20

* 4.4.2.4:

Delete major defect 141.

4.4.2.4: Major 147: Under Examination or Test, delete "&".

Add the following major defect:

"148	Functional runout of body cavity	0.40%	3.2	Gage"

* 4.4.2.4:

Add "(a)" after defect category 147.

Add the following major defect:

"149	(b) OD of bourrelets, max.	100%	3.1	Gage"

* Add the following Major defect:

"150	Band seat visible in rotating band cannelures	100%	3.3.2	Visual".

PAGE 21

* 4.4.2.4: Add the following notes to the notes block:

"(a) This inspection is not required when the body and ogive are supplied as an assembled unit.

(b) This inspection is not required when the bourrelet diameter on the Projectile Metal Parts (after paint) is inspected 100%."

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PAGE 23

* Delete NOTE B and substitute the following:

"(b) Sample size shall be as specified in 4.4.3.3.1 herein."

4.4.2.6:

Major 119: Delete in its entirety.

Major 121: Delete "Gage" and substitute "Visual".

PAGE 25

4.4.2.6: Add the following major examination:

"127 Taper incorrect 0.40% 3.1 Gage"

PAGE 26

4.4.2.7: Delete major category 103 in its entirety.

PAGE 28

4.4.2.7:

Minor 201, under Examination or Test, add "diameter" after "basic".

Minor 203, under Examination or Test, delete "Diameter on taper" and substitute "Taper incorrect".

Minors 204 and 205, delete "Gage" and substitute "Visual".

PAGE 29

* 4.4.2.8: Base and adapter assembly.

Critical 1: Add "After assembly and prior to painting."

Major 109: Add "After assembly and prior to painting."

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4.4.2.8: Major 104: Delete "Gage" and substitute "Visual".

4.4.2.8: Delete Major 104, 106 and 107 in their entirety.

PAGE 31

* 4.4.2.9: Major category

102: Add "(c)" after 100% in AQL column.

Add the following note to the notes block:

"(c) An option AQL of 0.40% applies when this defect is inspected 100% on the body before painting (see para 4.4.2.4, major defect 149)."

4.4.2.9: Add the following major categories:

"107	Lifting plug or gasket missing	-	100%	3.1	Visual
108	Lift plug detorque	-	0.40%	3.1	Gage"

4.4.2.9: Minor category

208: Delete "& fuze".

PAGE 33

4.4.2.10: Major category Delete "None defined" and add the following:

"101	Support cover missing (8 places) when applicable	100%	3.1	Visual".

PAGES 33a and 33b

* 4.4.2.11: Add this paragraph (2 sheets attached).

PAGE 33c

* 4.4.2.12: Add this paragraph (1 sheet attached).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9378453
4.4.2.11	Base (prior to prime and paint)				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special 1	Heat treatment verification		100%	3.6.1	4.5.1.3
Major 101	Metal defective		100%	3.5.1	4.5.4.2 (UT)
102	Yield strength and percent elongation		(a)	3.6	4.5.1.7
103	Length from shoulder to undercut at aft end of thread		0.40%	3.1	Gage
104	Depth to basic diameter in cavity radius.		0.40%	3.1	Gage
105	Length from shoulder to forward end		0.40%	3.1	Gage
106	Pitch diameter of thread, min.		0.40%	3.1	Gage
107	Major diameter of thread, min.		0.40%	3.1	Gage
108	Diameter of O-ring groove		0.40%	3.1	Gage
109	True position of O-ring groove		0.40%	3.1	Gage
110	Width of O-ring groove		0.40%	3.1	Gage
111	Diameter at forward edge of thread		0.40%	3.1	Gage
112	Angle of thread lead and follow chamfer, max.		0.40%	3.1	Gage
113	Surface finish of O-ring groove and shoulder		0.40%	3.1	Visual/Comparison Blocks
113	Straightness of shoulder		0.40%	3.1	Gage

NOTES: (a) 4.5.1.6

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9378453
4.4.2.11	Base (prior to prime and paint)				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Length from shoulder to basic boattail diameters (2 places)		0.40%	3.1	Gage
116	Runout of boattail (2 places)		0.40%	3.1	Gage
117	Thickness of radius webb		0.40%	3.1	Gage
118	Length of boattail		0.40%	3.1	Gage
Minor					
201	Inside diameter of boattail		0.65%	3.1	Gage
202	Runout of boattail inside diameter		0.65%	3.1	Gage
203	Boattail-webb radius		0.65%	3.1	Gage
204	Width of wrench slots, max.		0.65%	3.1	Gage
205	Depth of wrench slots, max.		0.65%	3.1	Gage
206	True position of wrench slots		0.65%	3.1	Gage
207	Any feature or operation missing		0.65%	3.1	Gage
208	Surface finish incorrect, except O-ring groove and shoulder		0.65%	3.1	Visual/Comparison Blocks
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9378453
4.4.2.12	Base				NEXT HIGHER ASSEMBLY 9352634
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major					
101	Threads damaged		100%	3.1	Visual
102	Salt spray		(a)	3.1	4.5.10
Minor					
201	Protective coating missing, incomplete or damaged		0.65%	3.11	Visual
202	Paint on threads, O-ring groove or shoulder		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) Sample size shall be specified in 4.4.3.3.1.

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PAGE 34

* 4.4.3.3.1: Delete in its entirety and substitute the following:

"4.4.3.3.1 Base (9352629) and ogive. First article bases and ogives shall withstand the salt spray test as prescribed in paragraph 3.10.1.2 of MIL-A-8625. Production lot bases and ogives shall comply with paragraph 3.10.1.2 of MIL-A-8625 except exposure shall be for 72 hours and show no more than 3 corrosion spots and/or pits on the significant surface. The significant surface on the ogive as prescribed in MIL-A-8625 may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto the ogive."

* 4.4.3.3.3: Delete "XM825" and substitute "M825".

Add paragraph 4.4.3.3.4 as follows:

" 4.4.3.3.4 Ogive. First article ogives shall withstand the salt spray test as prescribed in para 3.10.1.2 of MIL-A-8625. Production lot ogives shall comply with para 3.10.1.2 of MIL-A-8625 except exposure shall be for 72 hours and show no more than 3 corrosion spots and/or pits on the significant surface. the significant surface as prescribed in MIL-A-8625 may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto the ogive."

PAGE 35

* 4.5.1.2: Add the following to the end of paragraph a:

"Manual grinding is not permitted."

PAGE 36

4.5.1.3: Delete and substitute the following:

"4.5.1.3 Verification of heat treat of base. The test shall be made using procedures as prescribed in ASTM E-18. The readings shall be taken on the surface indicated on the applicable drawing."

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* Delete paragraph 4.5.1.5 in its entirety and substitute:

"4.5.1.5 Yield strength and percent elongation of ogive. The sample size for the tension test shall be one component from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The test specimens shall be taken from the midwall at a location where the gage length center of the specimen is approximately 1 inch from the finished rear face of the body thread end of the ogive, 180 apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Percent elongation shall be recorded to the nearest tenth percent. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample."

* Add 4.5.1.7 as follows:

"4.5.1.7 Hardness (base, 9378453). In order to determine uniformity of heat treatment, each base after heat treatment and before final machining shall be subjected to a hardness test with a Brinell Hardness Tester or any other method approved by the procuring activity. The test shall be performed in accordance with ASTM Method E-103 and the following:

a. The surface of the base in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. Flats and impressions shall be completely removed by final machining. manual grinding is not permitted.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

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d. The hardness test shall use a standard 10MM Brinell ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the base to be tested with a 10,000 -500 lb. clamping load. Second, apply a minor load of approximately 10% of the 3000 kg of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetrator travel from initial minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the base."

* Add paragraph 4.5.1.8 as follows:

"4.5.1.8 Yield strength and percent elongation of base. One of the hardest and one of the softest bases from each standard heat treatment lot or two of the hardest and two of the softest bases from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.7 shall be subjected to this test. Samples for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. From each base selected, one test specimen shall be taken transversely from near the center of the base webb, and two specimens shall be taken longitudinally 180 apart from the boattail forward corner area. The specimens shall conform to the largest obtainable round type diameter specimen as prescribed by ASTM Method E-8. If a .113 inch diameter cannot be machined from the longitudinal area of the base heat treat configuration, one rough forged base of the same mill heat of steel shall be included in each heat treatment lot and two longitudinal specimens shall therefrom be machined. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any longitudinal specimen to comply with the yield strength and elongation requirements, or failure of any transverse specimen to comply with the yield strength requirements of the applicable drawing shall be cause for rejection of the heat treatment lot. At the option of the contractor, the entire lot may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bases falling within this range will be accepted. All other bases will be rejected. Rejected bases may be reheat treated and retested."

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* 4.5.3.1: Delete paragraph and substitute the following:

"4.5.3.1: Inspection of band seat.

4.5.3.1.1 Welder qualification sampling. Initially the first five banded bodies after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the tests outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat metal (steel) (see 6.7). These bodies shall be subjected to the following tests:

a. Visual examination for compliance with 3.3.1.1 minimum melt requirement.

b. Visual examination for compliance with 3.5.2 metal defects, using the technique described in 4.5.3.1.3.

c. Turn down band seat to the diameter specified in 3.3.1.2 and again visually inspect in accordance with 4.5.3.1.3. Any defect indication per 3.3.1.2 at this depth is a reject.

d. Should any sample for any of the above tests fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.5.3.1.2 Post qualification sampling.

4.5.3.1.2.1 Etched band seat examination. Reduced sampling of one in each 8 hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable (see 6.9). One body per welder will be randomly selected and inspected for the following:

a. Visual examination for compliance with 3.3.1.1, minimum melting. The body selected for this test need not be the test sample having the lower rotating band iron content when tested in accordance with 4.5.5.

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b. Visual inspection for compliance with 3.5.2, metal defects, using the technique described in 4.5.3.1.3. If the sample fails to comply with 3.5.2 then the band seat will be turned down to the diameter specified in 3.3.1.2 and again visually inspected in accordance with 4.5.3.1.3.

c. Should any sample fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

d. Two consecutive failures when tested for a, or two consecutive failures when tested for b, for any welding machine will mandate the return to the original welder qualification sampling of paragraph 4.5.3.1.1. However, reduced sampling of one in each 8 hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.5.3.1.2.2 Sub-surface defect examination. The body having the highest acceptable iron content rotating band when tested per paragraph 4.5.5 during this 8 hour post qualification sampling shall be tested for compliance with 3.3.1.2. This body will have its band seat turned down to the diameter specified in paragraph 3.3.1.2 and visually inspected per paragraph 4.5.3.1.3. Failure of the body to comply with the specified requirements is cause for rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body shall also require that the body exhibiting the highest iron content from each welder be similarly tested. Failure of any body is cause for rejection of the quantity from each discrete welder represented by the sample. Two consecutive failures on any welding machine will mandate the return to the original welder qualification sampling of paragraph 4.4.3.1.1. However, reduced sampling of one in each 8 hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.5.3.1.3 Magnetic particle inspection. Magnetic particle inspection shall be as specified in MIL-I-6868 for wet fluorescent method with indirect magnetization by the continuous method, initially after inducing a circular magnetic field and again after inducing a longitudinal magnetic field.

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4.5.3.1.4 Retest. If the sample selected in 4.5.3.1.2.2 fails to comply with the requirements, then a minimum of 10% of the shell from the failed welder may be randomly selected for retest. These parts will be tested in accordance with 4.5.3.1.2.2. Only one retest per welder is allowed. Failure of any part will mandate the rejection of the product represented by the sample.

4.5.3.2: Add the following after the first sentence:

"At the option of the contractor, this test may be performed prior to machining of the rear counterbore, keyway/threads."

PAGE 39

* Delete paragraph 4.5.4 in its entirety and substitute the following:

"4.5.4 Metal defective test.

4.5.4.1 Base (9352629). The base shall be inspected for metal defects using the Eddy Current method. Bases failing to comply with the requirements of 3.5 shall be classed defective and removed from the lot. All bases passing this test shall be stamped in accordance with their applicable drawing. The standards and test method shall be in accordance with drawings 9280354 and 9280355.

4.5.4.2 Base (9378453). Every base shall be ultrasonically inspected for metal defects. Standards and test methods shall be in accordance with drawings 9280427 and 9280428 and with MIL-A-70625. Stamp letter "U" after acceptance as indicated on the base drawing.

4.5.4.2.1 Calibration of ultrasonic equipment. As a minimum the adequacy of the inspection process shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as four (4) intervals during production. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected."

4.5.5: Delete the next to last sentence.

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PAGE 40

Delete paragraph 4.5.7 in its entirety and substitute the following:

"4.5.7 Microstructure examination (ogive) and base, (9352629). One ogive and one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous network form shall be cause for rejection of the lot represented by the sample. Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys."

PAGE 44

Delete title, "Figure 2 Unacceptable Microstructures (250X)"

Delete "a" and substitute "Figure 2. Marginally Acceptable Microstructure (250X)"

Delete "b" and substitute "Figure 3. Unacceptable Microstructure (250X)"

PAGE 45

5.1.1, 5.1.2, and 5.1.3: Delete in their entirety.

PAGE 46

5.2: Delete and substitute the following:

"5.2 Packaging, packing and marking."

5.2.2: Delete and substitute the following:

"5.2.2 Packaging, Packing and Marking of Bases shall comply with the following:"

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Add the following paragraphs after paragraph 5.2.2:

5.2.2.1 Overseas shipments. Bases shall be semi-bulk or single item packaged, moisture proof packaged (barrier bag only), packed with cavity facing downward, marked, and palletized in accordance with MIL-STD-1169, Level A. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward.

5.2.2.2 Domestic shipments. Bases shall be semi-bulk or single item packaged, packed with the cavity facing downward, and marked in accordance with MIL-STD-1169, Level B. Bases shall be separated or partitioned (if bulk) in the unit package to protect the paint surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward. Palletization is optional."

* 6.1: Intended use: Change "XM825" to "M825/M825A1".

* 6.3 Delete "ARRADCOM, Attn: DRDAR-QAR-I, Dover, NJ 07801" and substitute "AMCCOM, Attn: AMSMC-QAR-I(D), Picatinny Arsenal, NJ 07806-5000."

* 6.4 Delete "ARRADCOM, Attn: DRDAR-QAR-Q, Dover, NJ 07801" and substitute "AMCCOM, Attn: AMSMC-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000."

* 6.5 Delete the last two lines, and substitute:

"AMCCOM, Attn: AMSMC-QAR-Q(D), Picatinny Arsenal, NJ 07806-5000 and PMCAWS, Attn: SMCAR-CCH-P, Picatinny Arsenal, NJ 07806-5000."

6.6 Delete "XM825" and substitute "M825".

* 6.9 Delete "ARRADCOM, Attn: DRDAR-LCU-M, Dover, NJ 07801" and substitute "ARDEC, Attn: SMCAR-CCH-P, Picatinny Arsenal, NJ 07806-5000."

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The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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